Project Title

Machine Downtime Analysis Dashboard (Power BI + Excel)

Bullet Points

- Built an interactive Power BI dashboard to analyse machine downtime using operational sensor data
- Performed data cleaning and feature engineering in Excel, including creation of risk scoring variables
- Designed KPIs including failure rate (51.7%), high-risk machine count, and operational distribution
- Developed visualisations to identify failure patterns across assembly lines and risk categories
- Conducted sensor-based analysis (torque, pressure, temperature) to investigate root causes of machine failure
- Identified that failures are driven by multi-factor conditions rather than single sensor thresholds
- Implemented slicers and dynamic filtering for interactive analysis across assembly lines

Objective

- Analyse machine downtime data to identify patterns, risk factors, and potential drivers of failure.

Approach

- Cleaned and structured raw sensor data in Excel
- Created calculated risk indicators (pressure, temperature, torque)
- Built correlation understanding between sensors and failure states
- Developed an interactive Power BI dashboard with KPIs and drill-down capability

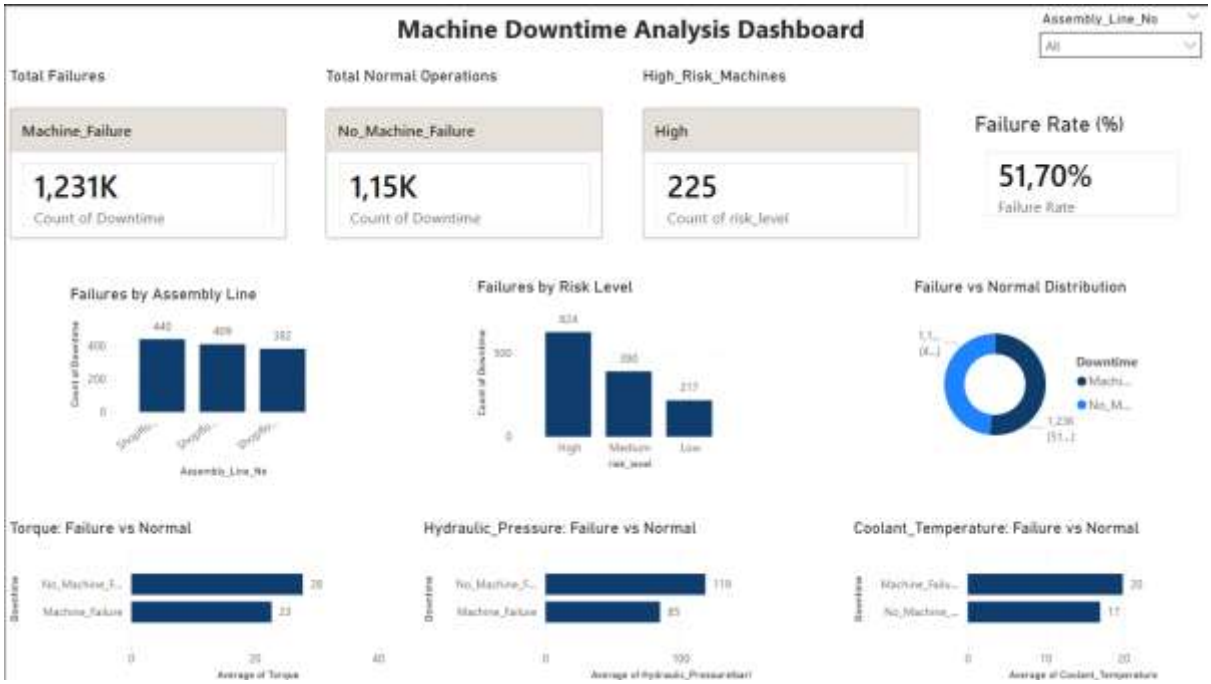
Key Insights

- Failure rate is **51.7%**, indicating high operational instability
- High-risk machines contribute the majority of failures
- Shopfloor-L1 shows the highest failure concentration
- Sensor variables (torque, pressure, temperature) do not independently explain failures

Conclusion

- Machine failures are likely driven by **combined sensor conditions**, suggesting the need for multi-variable monitoring systems rather than single-threshold alerts.

Tools

- Power BI
- Excel

Dashboard